# Ark Gupta

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## Technical Skills

**Languages:** C, C++, Python, SQL, JavaScript
**Frameworks & Backend:** React.js, Next.js, Node.js, FastAPI, Flask, REST APIs, Tailwind CSS
**Machine Learning & AI:** Scikit-learn, TensorFlow, PyTorch, sentence-transformers, FAISS, Keras, NLTK, spaCy
**Data & Visualization:** Pandas, NumPy, Matplotlib, Seaborn
**Databases:** MySQL, MongoDB, Firebase
**Tools:** Git, GitHub, Streamlit, Playwright, Jupyter

## Projects

**Zenness: AI-Powered Mental Health Assessment Platform** [Live] *Next.js, FastAPI, SQLAlchemy, JWT, Scikit-learn*

- Built and deployed a full-stack mental wellness platform using **Next.js** and **FastAPI** that predicts user stress levels through a **22-feature SVM-based ML pipeline** with secure **JWT authentication** and personalized wellness recommendations.
- Trained and optimized an **SVM classifier** on ∼1,100 assessment samples, achieving **91.89% accuracy** compared to a **34.24% baseline model** through behavioral and health-risk feature engineering.
- Developed scalable **REST APIs** and interactive dashboards for historical assessment tracking and longitudinal stress analysis using **SQLAlchemy** and **SQLite**; deployed on **Vercel** and **Render** with ∼262 ms average inference latency.

**EduPilot: AI-Powered Course Assistant** *Playwright, FAISS, sentence-transformers, Groq API, Streamlit*

- Secured **2nd Runner Up** at Masters' Union Hackathon, **XPECTO'26, IIT Mandi** — built an end-to-end **RAG pipeline** crawling **148 program pages** across **15+ institutions**, delivering source-attributed answers on curriculum, fees, and admissions for any program URL without redeployment.
- Engineered Playwright-based scraping of JS-rendered academic sites, chunking content into **1,443 semantic segments** (sliding-window, 40-token overlap) and indexing via **FAISS** with **all-MiniLM-L6-v2** embeddings achieving **0.16ms average retrieval latency**.
- Constrained **Llama3 (Groq API)** strictly to top-8 retrieved chunks via structured system prompting, eliminating hallucinations on out-of-context queries and enabling zero-shot generalization to unseen institutions.

## Open Source Contributions

**JuliaLang Ecosystem (JuliaLang, DataFrames.jl)** *Dec 2025 – Present*

- Refactored the README architecture for **JuliaLang/julia** (48k⋆, #60443), introducing a dedicated **"Contributing"** section that streamlines developer onboarding and contribution workflows – **merged** into master.
- Fixed invalid indexing behavior in **DataFrames.jl** (#3522) for **SubDataFrame** transformations with `copycols=false` by implementing error handling in selection logic, iterated on feedback from core maintainers.

## Education

**Indian Institute of Information Technology, Nagpur** Aug 2023 – Present
B.Tech in Computer Science & Engineering (Data Science and Analytics)

**Immaculate Conception Convent School, Lucknow** 2020 – 2022
Senior Secondary (Class XII - ICSE) **94%**

## Achievements & Certifications

- **2nd Runner-Up, Masters' Union Hackathon (XPECTO'26, IIT Mandi)**
- **Competitive Programming:** Solved 250+ DSA problems on **LeetCode**; peak rating **1555**.
- **Oracle Cloud Infrastructure (OCI) AI Foundations Associate** – Oracle *Feb 2026*
- **Postman API Fundamentals Student Expert** – Postman *Mar 2026*
- **Fundamentals of Deep Learning** – NVIDIA *Mar 2026*

## Relevant Coursework

**Core CS:** Data Structures and Algorithms, Object-Oriented Programming (OOP), Operating Systems, Database Management Systems (DBMS), Computer Networks, Foundations of Computing

**AI & Data Science:** Artificial Intelligence, Machine Learning, Deep Learning, Data Visualization, Big Data Analytics